DGMS 2019 • SECTION M: OPTICAL REMOTE SENSING & OPTICS

Question 92: Optics & Remote Sensing: Polarization of Reflected Light & Brewster's Law

Official Examination Guide • Plane Polarized Light, Specular Water Reflection & Rayleigh Scattering Polarization

OFFICIAL EXAMINATION QUESTION

Consider the following statements regarding reflected light:

1. The reflected light, vibrating in only one plane, is called polarized light.

2. The light reflected from a water body into the field of view of a camera is polarized light.

3. The light from a blue sky is non-polarized because it is reflected from non-metallic particles in the atmosphere.

Which of the statements given above is/are correct?

(a) 1 and 2 CORRECT ANSWER	(b) 2 and 3
(c) 3 only	(d) 1 only

1. Executive Summary & Core Principle

Light is a transverse electromagnetic wave where the electric field vector $ \mathbf{E}$ vibrates perpendicularly to the direction of propagation (Hecht, *Optics*; Lillesand & Kiefer, *Remote Sensing and Image Interpretation*):

- **Statement 1 is TRUE:** In unpolarized light, the electric field vector vibrates randomly in all possible planes perpendicular to the propagation ray. When the electric field vibrations are confined to a single fixed plane, the light is termed **linearly (or plane) polarized light**.
- **Statement 2 is TRUE:** When unpolarized sunlight reflects off a smooth non-metallic dielectric surface such as a water body ($n \approx 1.333$), the reflected ray is partially to completely polarized. According to **Brewster's Law**:

  > $\tan(\theta_B) = n_2 / n_1 \approx 1.333 \rightarrow \theta_B \approx 53.1^\circ$

  At the Brewster angle of incidence ($\theta_B \approx 53^\circ$), the component of the electric vector oscillating in the plane of incidence (p-polarization) is completely refracted into the water, and the reflected light is **100% plane-polarized parallel to the water surface** (s-polarization / horizontal polarization). This reflected polarized light enters camera lenses as blinding glare, which aerial photographers and remote sensing sensors suppress using polarizing filters.
- **Statement 3 is FALSE:** Diffuse blue skylight is produced by **Rayleigh scattering** of unpolarized sunlight by atmospheric gas molecules (N_2, O_2). Because molecular dipole oscillation does not radiate energy along the dipole axis, skylight scattered at an angle of 90° relative to the incident solar rays is **strongly linearly polarized** (reaching 70% to 85% polarization under clear skies). Stating that clear blue skylight is "non-polarized" contradicts basic physics!

Therefore, Statements **1 and 2** are correct, making Option **(a) 1 and 2** the correct answer.

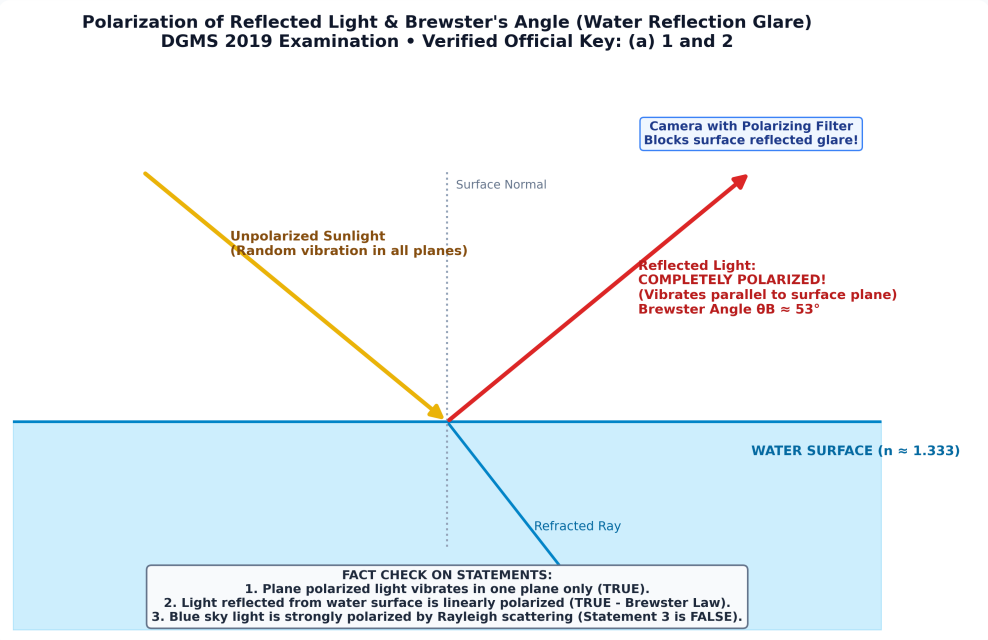


Figure 92.1: Reflection of unpolarized light from a water surface at Brewster's angle resulting in completely polarized reflected glare, contrasted with Rayleigh atmospheric scattering polarization (Grounded in Hecht Optics and Lillesand & Kiefer).

2. Mathematical Physics of Polarization by Reflection (Fresnel Equations)

The reflection coefficients for light polarized parallel ($r_\parallel$) and perpendicular ($r_\perp$) to the plane of incidence are given by Fresnel's equations:

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r∥ = tan(θi − θt) / tan(θi + θt)
r⊥ = − sin(θi − θt) / sin(θi + θt)
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When the angle between the reflected ray and the refracted ray is exactly 90° ($\theta_i + \theta_t = 90^\circ$), the denominator $\tan(\theta_i + \theta_t) \rightarrow \infty$, causing $r_\parallel = 0$. Consequently, the reflected beam contains *only* $r_\perp$ vibrations (vibrating parallel to the reflecting surface), resulting in **complete linear polarization**.

3. Evaluation of Options

- **Statement 1: CORRECT.** Defines linearly plane-polarized light.
- **Statement 2: CORRECT.** Reflection from water surfaces is linearly polarized (horizontal glare) according to Brewster's law.
- **Statement 3: INCORRECT.** Blue skylight is strongly polarized due to Rayleigh scattering at 90° from the Sun.

> **Key Takeaway:** Light vibrating in one plane = polarized (1); Water reflection = polarized glare via Brewster's Law (2); Blue skylight = polarized by Rayleigh scattering (3 is false). Hence, **(a) 1 and 2** is the correct answer.

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